EX. Let $\sigma \in Hom(V,W)$, Let σ^* be the dual map. Show that

① σ is injective $\Leftrightarrow$ σ^* is surjective

② σ is surjective $\Leftrightarrow$ σ^* is injective

Recall $\sigma: V \longrightarrow W$

inducing $\Downarrow$

dual map: $\sigma^*: W^* \to V^*$

$f \longmapsto f\sigma$

warning.

1. Not restricted to f.d.

2. comparing with f.d.

vocabulary

σ injection $\longleftrightarrow$ $\tau\sigma = 1_V \Leftrightarrow \sigma^* \tau^* = 1_{V^*}$

Rule $(AB)^T = B^T A^T$

$\Downarrow$

$(\tau\sigma)^* = \sigma^* \tau^*$

CH V Eigen- Theory

Recall $\sigma = \frac{d}{dx} \; C^\infty \longrightarrow C^\infty$

Fixed point $f \longmapsto \sigma(f) = 1 \cdot f \Leftrightarrow f = c \cdot e^x$, with f' written over $\sigma(f)$

$\Downarrow$

eigen value $f \longmapsto \lambda f = \sigma(f) = f' \Leftrightarrow f = ce^{\lambda x}$

Aim $\overset{\alpha_1 \cdots \alpha_n}{\hat{V}} \xrightarrow{\sigma} \hat{V}$

$v \overset{A}{\longmapsto} \sigma(v) = \lambda v = \lambda(\alpha_1, \cdots \alpha_n) X$

$v = (\alpha_1, \cdots \alpha_n) X$, $\sigma(v) = (\alpha_1, \cdots \alpha_n) AX$

so $(\alpha_1, \cdots \alpha_n)[AX] = (\alpha_1, \cdots \alpha_n)(\lambda X) \Rightarrow AX = \lambda X$

Thm Let $V/\mathbb{F}$ be v.s. Then can define eigen-theory for $\forall \sigma \in End V$ ad hoc, if $\dim V = n$. Then the eigen-theory over V is exactly that of matrices.

That is, $\forall \sigma \in End V$, its eigenvalues are exactly the eigenvalues of A, for any A which is the matrix of σ.

Conclusion $\dim V = n < +\infty$ $\tau \in \mathrm{End}V$.

Fix basis of V $\{\alpha_1, \cdots \alpha_n\} = \mathcal{B}$

Let A be the matrix of τ under $\mathcal{B}$

Then. i) $\sigma(\tau) = \sigma(A) \in \mathbb{C}$

ii) Let $\lambda \in \sigma(A)$ and $AX = \lambda X$, $\overset{\in F^n}{X \neq 0}$

Then $\tau((\alpha_1 \cdots \alpha_n) X) = \lambda(\alpha_1, \cdots \alpha_n) X$

Next question Find a "good" algorithm for $\sigma(\tau) \rightsquigarrow \sigma(A) = \{\lambda \in \mathbb{C} \mid \det(\lambda I - A) = 0\}$

Schur's Lemma: $U^* A U = T$ upper triang.)

so $\sigma(A) = \sigma(T) = \{t_{11}, \cdots t_{nn}\}$

Recall Let $\tau \xleftrightarrow{\alpha_1 \cdots \alpha_n} A$ $\qquad \tau \underset{(\alpha_1 \cdots \alpha_n)P}{\xleftrightarrow{\beta_1 \cdots \beta_n}} B = P^{-1}AP$

$\tau((\alpha_1 \cdots \alpha_n)P) = (\alpha_1, \cdots \alpha_n)(AP) = (\alpha_1, \cdots \alpha_n) PB$.

Thm 2 $\forall \sigma \in \mathrm{End}V$ Fix a basis of V $\{\alpha_1, \cdots \alpha_n\} = \mathcal{B}$

Let A be the matrix of σ under $\mathcal{B}$. Then the matrix of σ under $(\alpha_1, \cdots \alpha_n)P$ is $P^{-1}AP$ ad hoc. if $\mathbb{F} = \mathbb{C}$, then $\exists$ basis $v_1, \cdots v_n$ of V s.t. the matrix of σ under $\{v_1, \cdots v_n\}$ is upper triangular.

Ex ① $\sigma: \overset{3}{V} \longrightarrow \overset{3}{V} = \{y \mid y''' + y'' + y' + y = 0\}$

Find v_1, v_2, v_3:

$\sigma: v_1 \longmapsto a_{11} v_1$

$v_2 \longmapsto a_{12} v_1 + a_{22} v_2$

$v_3 \longmapsto a_{13} v_1 + a_{23} v_2 + a_{33} v_3$

$\sigma(v_1, v_2, v_3) = (v_1, v_2, v_3)\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{pmatrix}$

② $\sigma: \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ rotation $\frac{\pi}{2}$

$\begin{pmatrix} x \\ y \end{pmatrix} \longmapsto \begin{pmatrix} -y \\ x \end{pmatrix}$ $\quad A: \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ 不能上三角化

RK 1. If $\mathbb{F} \neq \mathbb{C}$, triangularization fails in general

2. Find sufficient and necessary conditions for diagonalization.

Recall $A \in \mathbb{C}^{n\times n}$ is diagonalizable $\Leftrightarrow$ $(m_A'(x), m_A(x)) = 1$ $\Leftrightarrow$ $\mathbb{C}^n = \sum_{\lambda\in\sigma(A)} \oplus V_\lambda$

$\Leftrightarrow$ A has a complete system of eigenvectors

Thm Let $\tau \in \operatorname{End} V$ (f.d.). Then τ is diagonalizable $\Leftrightarrow$ τ has a complete system of eigenvectors $\Leftrightarrow$ $V = \sum \oplus V_\lambda$, where $V_\lambda = \{v \in V \mid \sigma(v) = \lambda v\}$ $\lambda \in \sigma(\tau)$

Ex. ① $\tau = \frac{d}{dx}$ $F[x]_2 \to F[x]_2$ $\sigma(\tau) = \{0\}$

② $\rho = x\frac{d}{dx}$ $\rho f = \lambda f$ i.e. $xf' = \lambda f$

$\int \frac{f'}{f} dx = \lambda \int \frac{1}{x} dx \Rightarrow \ln f = \ln x^\lambda + C \Rightarrow f = C x^\lambda$

$\lambda = 0, 1, 2 \Rightarrow$ diagonalizable

☆In one word, everything true for matrices is true for $\tau \in \operatorname{End} V$

So can define $\det \tau$, $\operatorname{tr} \tau$, $\det(\lambda I - \tau)$

Cor. Suppose $\tau \in \operatorname{End} V$ is diagonalizable, say $V = \sum_{\lambda\in\sigma(\tau)} \oplus V_\lambda$. Then ① $\tau = \sum_{\lambda\in\sigma(\tau)} \oplus \tau_\lambda$,

where $\tau_\lambda \in \operatorname{End} V_\lambda$: $V_\lambda \to V_\lambda$ and

$v \mapsto \lambda v$

$\tau_{\lambda_1} \oplus \cdots \oplus \tau_{\lambda_n}$: $V_{\lambda_1} \oplus \cdots \oplus V_{\lambda_n} \longrightarrow V_{\lambda_1} \oplus \cdots \oplus V_{\lambda_n}$

$v_1 + \cdots + v_n \longmapsto \lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_n v_n$

② $\tau = \sum_{\lambda\in\sigma(\tau)} \tau|_{V_\lambda}$ where $\tau|_{V_\lambda}$ $V \longrightarrow V$

$v \longmapsto \tau|_{V_\lambda}(v) = \tau(v_\lambda)$

Ex. ① τ $\mathbb{R}^3 \longrightarrow \mathbb{R}^3$

$\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ $\begin{pmatrix} x+y+z \\ x+y+z \\ x+y+z \end{pmatrix}$

$\sigma(\tau) = \{3, 0, 0\}$ $V_3 = \mathbb{F}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ $V_0 = \mathbb{F}\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \oplus \mathbb{F}\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$

$\tau_3 : V_3 \to V_3$

$\begin{pmatrix} x \\ x \\ x \end{pmatrix} \to 3\begin{pmatrix} x \\ x \\ x \end{pmatrix}$

$\tau_0 : V_0 \mapsto 0$

$\tau = \tau_3 \oplus \tau_0$

$\tau\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \tau\left[a\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + b\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + c\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}\right]$

$\tau|_{V_3}(v) = \tau|_{V_3}(v_3 + v_0) = \tau(v_3)$